

# American Made Physics: Spectral Gap Control, Curvature Energy, and Coherent Transport

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## Abstract

We develop a constructive operator-theoretic framework in which system behavior is controlled via spectral gap enforcement. For a class of self-adjoint generators with compact resolvent, we establish explicit gap amplification, exponential suppression of excited modes, entropy collapse, and consequences for information concentration and force extraction through boundary stress. The results are formulated with explicit constants and uniform bounds.

## 1 Operator Framework

**Definition 1.1** (Base Operator). Let  $\Omega \subset \mathbb{R}^d$  be bounded with  $C^{1,1}$  boundary. Define

$$A_0 := -\Delta_D + \mu_0 I + V,$$

on  $L^2(\Omega)$ , where:

- $\Delta_D$  is the Dirichlet Laplacian,
- $\mu_0 > 0$ ,
- $V \in L^\infty(\Omega; \mathbb{R})$ .

**Proposition 1.2** (Self-Adjointness and Compact Resolvent).  $A_0$  is self-adjoint with compact resolvent. Its spectrum is discrete:

$$\text{Spec}(A_0) = \{\lambda_n\}_{n=0}^\infty, \quad \lambda_n \rightarrow \infty.$$

*Proof.* The Dirichlet Laplacian has compact resolvent. Since  $V$  is bounded, Kato–Rellich implies self-adjointness is preserved. Compactness of the resolvent follows from relative boundedness.  $\square$

**Definition 1.3** (Protected-Mode Penalization). Let  $\phi_0$  be a normalized ground state of  $A_0$ . Define

$$P_\perp := I - |\phi_0\rangle\langle\phi_0|, \quad A^{(\mu)} := A_0 + \mu P_\perp, \quad \mu > 0.$$

## 2 Spectral Gap Control

**Theorem 2.1** (Gap Amplification). *Let  $\Delta^{(0)} := \lambda_1^{(0)} - \lambda_0^{(0)}$ . Then*

$$\Delta^{(\mu)} \geq \Delta^{(0)} + \mu.$$

*Proof.* For any  $\psi \perp \phi_0$ ,

$$\langle \psi, A^{(\mu)} \psi \rangle = \langle \psi, A_0 \psi \rangle + \mu \|\psi\|^2 \geq (\lambda_1^{(0)} + \mu) \|\psi\|^2.$$

Thus  $\lambda_1^{(\mu)} \geq \lambda_1^{(0)} + \mu$ , while  $\lambda_0^{(\mu)} = \lambda_0^{(0)}$ . □

**Corollary 2.2** (Spectral Barrier). *Define*

$$c_* := \lambda_1^{(0)} + \mu - \lambda_0^{(0)}.$$

*Then*

$$\inf \text{Spec}(A^{(\mu)}) = \lambda_0^{(0)}, \quad \text{Spec}(A^{(\mu)}) \setminus \{\lambda_0^{(0)}\} \subset [\lambda_0^{(0)} + c_*, \infty).$$

## 3 Semigroup and Resolvent Estimates

**Theorem 3.1** (Semigroup Decay). *For  $t \geq 0$ ,*

$$\|e^{-tA^{(\mu)}} P_\perp\| \leq e^{-c_* t}.$$

*Proof.* Spectral calculus yields

$$\|e^{-tA^{(\mu)}} P_\perp\| = \sup_{n \geq 1} e^{-t\lambda_n^{(\mu)}} \leq e^{-t(\lambda_0^{(0)} + c_*)}.$$

□

**Theorem 3.2** (Resolvent Bound). *For  $\Re z < \lambda_0^{(0)} + c_*$ ,*

$$\|(A^{(\mu)} - zI)^{-1} P_\perp\| \leq \frac{1}{c_* - \Re(z - \lambda_0^{(0)})}.$$

## 4 Thermal Asymptotics

Let

$$Z(\beta) := \text{Tr}(e^{-\beta A^{(\mu)}}).$$

**Theorem 4.1** (Partition Function Expansion).

$$Z(\beta) = e^{-\beta \lambda_0^{(0)}} (1 + R(\beta)),$$

*with remainder*

$$|R(\beta)| \leq C e^{-\beta c_*}.$$

*Proof.*

$$Z = e^{-\beta\lambda_0} + \sum_{n \geq 1} e^{-\beta\lambda_n} \leq e^{-\beta\lambda_0} \left( 1 + \sum_{n \geq 1} e^{-\beta c_*} \right).$$

□

**Definition 4.2** (Control Parameter).

$$\Xi := \beta c_*.$$

**Theorem 4.3** (Entropy Bound). *Let  $p_n = e^{-\beta\lambda_n}/Z$ . Then*

$$S = - \sum p_n \log p_n \leq C\Xi e^{-\Xi}.$$

## 5 Information-Theoretic Consequences

**Proposition 5.1** (Mode Suppression). *For  $n \geq 1$ ,*

$$p_n \leq C e^{-\Xi}.$$

**Proposition 5.2** (Fisher Information Scaling). *For a phase parameter  $\theta$  encoded in  $\phi_0$ ,*

$$\text{Var}(\delta\theta) \leq C e^{-\Xi}, \quad \mathcal{I}(\theta) \geq C^{-1} e^{\Xi}.$$

## 6 Curvature Energy and Boundary Force

Let  $A_\mu$  be a gauge connection with curvature  $F_{\mu\nu}$ .

**Definition 6.1** (Field Energy).

$$E_{\text{field}} := \int_{\Omega} \text{Tr}(F_{\mu\nu} F^{\mu\nu}) dV.$$

**Definition 6.2** (Stress Tensor Flux).

$$\mathbf{F}_{\text{net}} = \oint_{\partial\Omega} T \cdot n dA.$$

**Proposition 6.3** (Spectral Concentration of Force). *Under  $\Xi \gg 1$ , contributions to boundary flux from excited spectral sectors are exponentially suppressed:*

$$\mathbf{F}_{\text{net}} = \mathbf{F}_0 + O(e^{-\Xi}),$$

where  $\mathbf{F}_0$  is the contribution from the protected mode.

## 7 Conclusion

We have established that spectral gap enforcement yields:

- exponential suppression of unwanted modes,
- entropy collapse,
- information concentration,
- directional force extraction.

These results unify thermodynamic, informational, and mechanical behavior under a single control parameter  $\Xi$ .